João Dimas

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Education

Washington University in St. Louis
PhD in Finance, Olin Business School — 2022–2026
Advisor: Todd Gormley
Dissertation: The Economics of Technical Debt: Venture Capital, Cybersecurity, and Engineering Quality

Washington University in St. Louis
Doctoral studies in Economics; transferred to the PhD in Finance program. — 2020–2022

Université Paris I Panthéon-Sorbonne
MSc in Financial Economics — 2018

Universidade FUMEC
B.Sc. in Computer Science — 2009

Research Fields

Corporate finance and the economics of technology (AI, software quality, and cybersecurity).

Job Market Paper

Venture Capital and Engineering Quality: Evidence from Software Startups

with L. Pnevmatikos (third-year PhD student)

First Prize, Buchan Prize Paper Competition, Olin Business School.

We document a new dimension of venture capital (VC) professionalization: the way startups build and maintain software. We use monthly snapshots of public source code from 569 firms, covering 13.2 billion lines to estimate startups' technical debt—the dollar cost of correcting problems that make code harder to maintain. Following the first institutional VC entry, funded firms' critical code issues per line decline by 15% and technical debt per line by 12% relative to their pre-funding averages, compared with otherwise similar firms without VC backing. Moreover, engineers already active before VC financing become less likely to introduce bugs, and later funding rounds bring substantial hiring without detectable code quality gains. Descriptions accompanying 5.6 million code contributions reveal more systematic documentation and organization of development work after a VC's arrival. Together, the findings suggest that VC professionalization extends beyond governance and leadership to the engineering work of startups' existing teams.

Working Papers

AI and Technical Debt: Evidence from 4 Billion Lines of Code

with E. Lu and L. Pnevmatikos

We study whether generative AI helps firms build software assets while creating hidden liabilities in the form of future maintenance costs. Comparing projects around first detected AI use with projects where use has not yet been detected, we find that total issues, bugs, and technical debt per line decline by 1.4%, 2.1%, and 0.7% of their pre-detection averages. Quality gains concentrate in projects with weaker prior development practices, and the ratio of technical debt to software features declines relative to comparison projects.

Patch Now, Pay Later? Cybersecurity Vulnerabilities and Technical Debt

with L. Pnevmatikos

We study how cybersecurity vulnerabilities affect firms' growth and investment in software. Comparing firms exposed to vulnerability discoveries in third-party software with unaffected firms, we find approximately 10% slower headcount growth relative to its pre-treatment average and a decline in acquisition probability equivalent to 26% of its sample average. Development shifts toward repairs for twelve months, while feature-related activity declines at longer horizons. Technical debt and cognitive complexity accumulate faster, with growth increases equivalent to approximately 0.7% and 1.4% of their pre-treatment averages, respectively.

Work in Progress

Acquisition Prospects and Startup Strategy

with A. Jain and L. Pnevmatikos

We examine how a decline in acquisition prospects changes startups' product development and hiring, and whether it encourages investment in more durable standalone businesses.

Hedge or Speculation? Evidence from Brazil

with Gustavo Cortes, Sergio Leao, and Alexandre Pecora

We study how Brazilian non-financial firms adjust currency-derivative positions in response to market conditions and the past performance of their positions, distinguishing hedging from speculation.

Industry Papers

Toh, W. Q., W. Zheng, Y. Wang, and J. Dimas (2025). A White Box Framework for Estimating Long-Term Experimental Impact via Short-Term Metrics. ACM KDD 4th End to End Customer Journey Optimization Workshop.

We develop a framework for estimating long-term experimental effects from short-term metrics while making the underlying causal mechanisms explicit. The framework connects short-term measures to a long-term outcome through its key drivers, validates estimates using both pseudo-experiments and actual experiments, and addresses implementation on an experimentation platform.

Honors

First Prize, Buchan Prize Paper Competition, Olin Business School (with L. Pnevmatikos).

Conferences

12th Annual USC Marshall PhD Conference in Finance June 15, 2026

Job market paper, presented by L. Pnevmatikos.

Research and Professional Experience

Roblox

PhD Data Scientist Intern, Experimentation Platform 2025

- Developed methodology for estimating 365-day customer lifetime value treatment effects from 14-day experimental outcomes.
- Estimated engagement–monetization trade-offs and designed validation against realized long-run outcomes to inform experimentation-platform decisions.

Central Bank of Brazil (Banco Central do Brasil)

Economist, Research Department 2018–2021

- Contributed to research on inflation dynamics and monetary-policy transmission, producing forecasts and alternative macroeconomic scenarios to support policy decisions.
- Researched credit markets, banking efficiency, financial stability, and their effects on the real economy.
- Developed macroeconomic modeling and forecasting tools and reproducible empirical workflows; contributed analytical inputs to the Inflation Report and policy briefings.

Software Engineer, IT Department 2012–2016

CI&T

Software Engineer and Technical Lead 2008–2012

CEMIG

Software Engineer 2007–2008

Teaching Experience

Washington University in St. Louis

Teaching Assistant 2021–2025

PhD-Level Courses

Applied Econometrics	Fall 2021
Empirical Methods in Finance	Spring 2024
Introduction to Asset Pricing	Fall 2024

Other Graduate Courses

Advanced Corporate Finance I — Valuation	Spring 2024
Advanced Corporate Finance II — Financing	Fall 2022; Spring 2024
Derivative Securities	Fall 2023, 2024, 2025

Undergraduate Courses

Advanced Financial Management	Spring 2024, 2025
Computational Macroeconomics	Spring 2022
Intermediate Microeconomic Theory	Spring 2022
Options, Futures and Derivative Securities	Fall 2025

References

Olin Business School, Washington University in St. Louis

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